Problem Set 1.1 – 1.3: Verification Document

Focus: Subscriber Directory, Exchange Distribution, and Local Calls

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1. **Problem 1.1: Subscriber Directory Mapping**
   Produce a table consisting of the names and addresses of the subscribers and their phone numbers.

   Relational Algebra Expression

   $\pi_{name,address,area,office,number}(subscribers \bowtie_{portid} lines)$

   Solution:

```sql
SELECT
    s.name,
    s.address,
    l.area,
    l.office,
    l.number
FROM
    subscribers s
    JOIN lines l ON s.portid = l.portid;
```

2. **Problem 1.2: Area and Office Code Aggregation**
   Produce a table that lists all the area code and office code combinations in the database and the number of subscribers for each.

   Relational Algebra Expression

   $\gamma_{area,office;count(*)\rightarrow num_subscribers}(lines)$

   Solution:

```sql
SELECT
    area,
    office,
    COUNT(*) AS num_subscribers
FROM
    lines
GROUP BY
    area,
    office;
```

3. **Problem 1.3: Intra-Switch Local Call Detection**
   List the names of all those subscribers who are the originators of a call to someone who is also a subscriber on the same switch.

   Relational Algebra Expression

   $\pi_{name}(subscribers \bowtie_{portid=orig} (calls \bowtie_{orig=portid} \rho_{L1}(lines) \bowtie_{term=portid} \rho_{L2}(lines)))$

   Solution:

```sql
SELECT DISTINCT
    s.name
FROM
    calls c
    JOIN lines l1 ON c.orig = l1.portid
    JOIN lines l2 ON c.term = l2.portid
    JOIN subscribers s ON c.orig = s.portid;
```

End of Review Document